Mathematics Method for Physics–Lecture 18

Equations of Mathematical Physics and Travelling Waves

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We derive the basic equations of mathematical physics, state the data needed to select a solution, and solve the one-dimensional wave equation by characteristic coordinates. Throughout, $a > 0$ denotes wave speed; $\kappa > 0$ denotes thermal diffusivity, written as a^2 in the handwritten heat-equation notes.

1 Three Basic Equations

1.1 Wave, Heat, and Potential Equations

For a scalar field $u = u(\boldsymbol{x}, t)$, the Laplacian in three dimensions is

$$\Delta u = u_{xx} + u_{yy} + u_{zz}. \tag{1}$$

The three principal models considered in this course are:

1. **Wave equation:** displacement evolves according to
$$u_{tt} - a^2 \Delta u = f(\boldsymbol{x}, t). \tag{2}$$
2. **Heat equation:** temperature evolves according to
$$u_t - \kappa \Delta u = f(\boldsymbol{x}, t). \tag{3}$$
3. **Potential equation:** a stationary field satisfies
$$\Delta u = f(\boldsymbol{x}). \tag{4}$$

When the right-hand side is zero, the equation is **homogeneous**. The homogeneous potential equation is **Laplace's equation**; its solutions are called **harmonic functions**. The nonhomogeneous equation $\Delta u = f$ is a form of **Poisson's equation**. The source symbol f has a different physical meaning in each model.

1.2 Classification of Second-Order Equations

Consider a linear equation in two independent variables whose principal part is

$$Au_{xx} + 2Bu_{xy} + Cu_{yy} + \text{lower-order terms} = f. \tag{5}$$

At a point where A, B, C do not all vanish, its type is determined by $D = B^2 - AC$:

- **Hyperbolic:** $D > 0$, as in $u_{tt} - a^2 u_{xx} = 0$.
- **Elliptic:** $D < 0$, as in $u_{xx} + u_{yy} = 0$.
- **Parabolic:** $D = 0$, as in $u_t - \kappa u_{xx} = 0$.

The characteristic-direction equation is

$$A(\mathrm{d}y)^2 - 2B\,\mathrm{d}x\,\mathrm{d}y + C(\mathrm{d}x)^2 = 0. \tag{6}$$

Its discriminant is $4(B^2 - AC)$, which gives the same classification. If the coefficient of u_{xy} is instead denoted by B rather than $2B$, the corresponding discriminant is $B^2 - 4AC$.

2 Derivation of the Governing Equations

2.1 Small Transverse Vibrations of a String

Let a taut string have constant tension $T > 0$ and constant mass per unit length $\rho_s > 0$. Let $u(x,t)$ be its transverse displacement, and let $q(x,t)$ be the transverse force per unit length. Assume small slopes, $|u_x| \ll 1$, so that arc length is approximately horizontal length and $\sin\alpha \approx \tan\alpha = u_x$.

For the element $[x, x+\mathrm{d}x]$, Newton's second law gives, to first order,

$$T\big[u_x(x+\mathrm{d}x,t) - u_x(x,t)\big] + q(x,t)\,\mathrm{d}x = \rho_s\,\mathrm{d}x\,u_{tt}(x,t). \tag{7}$$

Divide by $\mathrm{d}x$ and take the limit:

$$\rho_s u_{tt} = T u_{xx} + q, \qquad u_{tt} - a^2 u_{xx} = f, \qquad a^2 = \frac{T}{\rho_s}, \quad f = \frac{q}{\rho_s}. \tag{8}$$

The case $q = 0$ describes free vibration; $q \neq 0$ describes forced vibration. The analogous membrane model in two spatial dimensions contains $u_{xx} + u_{yy}$ instead of u_{xx}.

2.2 Heat Conduction from Energy Conservation

Let $K > 0$ be thermal conductivity, $\rho > 0$ density, and $c_p > 0$ specific heat, all constant. Fourier's law states that the heat-flux vector is

$$\boldsymbol{q} = -K\nabla u. \tag{9}$$

Use the **outward** unit normal $\boldsymbol{n}$ on the boundary of an arbitrary fixed volume Ω. If Q is heat generated per unit volume per unit time, conservation of energy gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega \rho c_p u\,\mathrm{d}V = -\int_{\partial\Omega} \boldsymbol{q}\cdot\boldsymbol{n}\,\mathrm{d}S + \int_\Omega Q\,\mathrm{d}V. \tag{10}$$

Substitute Fourier's law and apply the divergence theorem:

$$\int_\Omega \rho c_p u_t\,\mathrm{d}V = \int_\Omega K\Delta u\,\mathrm{d}V + \int_\Omega Q\,\mathrm{d}V. \tag{11}$$

Because Ω is arbitrary, the local equation is

$$u_t - \kappa\Delta u = f, \qquad \kappa = \frac{K}{\rho c_p}, \quad f = \frac{Q}{\rho c_p}. \tag{12}$$

For spatially varying conductivity, the conduction term before division by ρc_p is $\nabla \cdot (K \nabla u)$ rather than $K \Delta u$.

2.3 Stationary Temperature and Potential Problems

When a temperature field becomes time independent, $u_t = 0$. With no internal heat source, the heat equation reduces to $\Delta u = 0$. Thus the same potential equation arises as the steady limit of heat conduction. With a source, the steady equation is $\Delta u = -Q/K$.

3 Initial and Boundary Conditions

3.1 Initial Data

A governing equation alone usually admits many solutions. A physical problem also specifies initial and/or boundary data.

- The heat equation is first order in time and requires the initial temperature $u(\boldsymbol{x}, 0) = \varphi(\boldsymbol{x})$.
- The wave equation is second order in time and requires both initial displacement and velocity:

$$u(\boldsymbol{x}, 0) = \varphi(\boldsymbol{x}), \qquad u_t(\boldsymbol{x}, 0) = \psi(\boldsymbol{x}). \tag{13}$$

A problem with initial data on an unbounded spatial domain is an **initial-value problem**. On a bounded domain, initial data and spatial boundary data together form an **initial-boundary-value problem**.

3.2 Three Types of Boundary Conditions

Let $\partial_n u = \nabla u \cdot \boldsymbol{n}$, with $\boldsymbol{n}$ outward. The standard boundary conditions are:

1. **Dirichlet condition:** $u|_{\partial\Omega} = g$ prescribes the field itself.
2. **Neumann condition:** $\partial_n u|_{\partial\Omega} = g$ prescribes its normal derivative.
3. **Robin condition:** $(\partial_n u + \sigma u)|_{\partial\Omega} = g$ prescribes a linear combination.

For heat conduction, a homogeneous Dirichlet condition fixes the boundary at zero reference temperature; a homogeneous Neumann condition describes insulation. Newton cooling into surroundings at temperature u_∞ gives

$$-K \partial_n u = h(u - u_\infty), \qquad \partial_n u + \frac{h}{K} u = \frac{h}{K} u_\infty, \tag{14}$$

where $h \geq 0$ is a heat-transfer coefficient.

For a vibrating string, $u = 0$ describes a fixed end and $u_x = 0$ a free end. An end attached to a linear spring of stiffness k_s satisfies $T \partial_n u + k_s u = 0$. On $0 < x < \ell$, the outward derivative is $-u_x$ at $x = 0$ and u_x at $x = \ell$, so spring conditions become

$$-T u_x(0, t) + k_s u(0, t) = 0, \qquad T u_x(\ell, t) + k_s u(\ell, t) = 0, \tag{15}$$

when the same spring stiffness is used at both ends. The sign depends on the endpoint and normal convention.

Compatibility. For a smooth fixed-end wave solution, the initial data must agree with the boundary conditions, for example $\varphi(0) = \varphi(\ell) = 0$ and $\psi(0) = \psi(\ell) = 0$ when these endpoint traces exist.

4 Travelling Waves and Characteristic Coordinates

4.1 Reduction to a Mixed Derivative

Consider the homogeneous wave equation on the whole line:

$$u_{tt} - a^2 u_{xx} = 0, \qquad -\infty < x < \infty. \tag{16}$$

Introduce characteristic coordinates and write the transformed unknown as U:

$$\xi = x + at, \qquad \eta = x - at, \qquad u(x,t) = U(\xi, \eta). \tag{17}$$

The chain rule yields

$$\begin{aligned} u_x &= U_\xi + U_\eta, & u_t &= a(U_\xi - U_\eta), \\ u_{xx} &= U_{\xi\xi} + 2U_{\xi\eta} + U_{\eta\eta}, & u_{tt} &= a^2(U_{\xi\xi} - 2U_{\xi\eta} + U_{\eta\eta}). \end{aligned} \tag{18}$$

Consequently,

$$u_{tt} - a^2 u_{xx} = -4a^2 U_{\xi\eta} = 0. \tag{19}$$

Integrating $U_{\xi\eta} = 0$ first in η and then in ξ gives $U = F(\xi) + G(\eta)$. Hence the general twice continuously differentiable solution is

$$\boxed{u(x,t) = F(x + at) + G(x - at).} \tag{20}$$

The term $F(x + at)$ travels to the left at speed a; $G(x - at)$ travels to the right. The lines $x \pm at =$ constant are the two characteristic families.

4.2 Derivation of d'Alembert's Formula

Impose initial data $u(x, 0) = \varphi(x)$ and $u_t(x, 0) = \psi(x)$. Then

$$F(x) + G(x) = \varphi(x), \qquad F'(x) - G'(x) = \frac{\psi(x)}{a}. \tag{21}$$

Let $H'(x) = \psi(x)$. Up to constants that cancel in $F + G$,

$$F(x) = \frac{\varphi(x)}{2} + \frac{H(x)}{2a}, \qquad G(x) = \frac{\varphi(x)}{2} - \frac{H(x)}{2a}. \tag{22}$$

Substitution produces **d'Alembert's formula**:

$$\boxed{u(x,t) = \frac{\varphi(x + at) + \varphi(x - at)}{2} + \frac{1}{2a} \int_{x-at}^{x+at} \psi(s)\, \mathrm{d}s.} \tag{23}$$

For example, $\varphi \in C^2$ and $\psi \in C^1$ are sufficient for this formula to define a classical solution. Its value at (x,t) depends only on initial data in $[x-at, x+at]$, expressing finite propagation speed.

Example: displacement x and velocity $\cos x$. With $\varphi(x) = x$ and $\psi(x) = \cos x$,

$$u(x,t) = x + \frac{\sin(x+at) - \sin(x-at)}{2a} = x + \frac{\cos x \sin(at)}{a}. \tag{24}$$

5 Nonhomogeneous Wave Equations

5.1 The Source-Term Extension

For $u_{tt} - a^2 u_{xx} = f(x,t)$ on the whole line, the extension of d'Alembert's formula is

$$\begin{aligned} u(x,t) = \frac{\varphi(x+at) + \varphi(x-at)}{2} + \frac{1}{2a}\int_{x-at}^{x+at} \psi(s)\,\mathrm{d}s \\ + \frac{1}{2a}\int_0^t \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(s,\tau)\,\mathrm{d}s\,\mathrm{d}\tau. \end{aligned} \tag{25}$$

The last term is the response to the source with zero initial data. It follows by applying the homogeneous initial-velocity formula to each forcing time τ and integrating those responses. Differentiation verifies that its wave operator is f and that its initial displacement and velocity vanish, under sufficient regularity.

5.2 Reduction by a Particular Solution

If a convenient particular solution W satisfies $W_{tt} - a^2 W_{xx} = f$, let $v = u - W$. Then v satisfies the homogeneous equation, with adjusted initial data

$$v(x,0) = \varphi(x) - W(x,0), \qquad v_t(x,0) = \psi(x) - W_t(x,0). \tag{26}$$

Example: a time-dependent force. Solve

$$\begin{cases} u_{tt} - u_{xx} = -\sin t, & x \in \mathbb{R},\ t > 0, \\ u(x,0) = \cos x, \qquad u_t(x,0) = \mathrm{e}^x. \end{cases} \tag{27}$$

Choose $W(t) = \sin t$. Then $v = u - \sin t$ has initial displacement $\cos x$ and initial velocity $\mathrm{e}^x - 1$. D'Alembert's formula with $a = 1$ gives

$$\begin{aligned} v(x,t) &= \frac{\cos(x+t) + \cos(x-t)}{2} + \frac{1}{2}\int_{x-t}^{x+t} (\mathrm{e}^s - 1)\,\mathrm{d}s \\ &= \cos x \cos t + \mathrm{e}^x \sinh t - t. \end{aligned} \tag{28}$$

Therefore,

$$\boxed{u(x,t) = \cos x \cos t + \mathrm{e}^x \sinh t - t + \sin t.} \tag{29}$$

Direct substitution checks the equation and both initial conditions. For a finite string, spatial boundary conditions must also be enforced; the next lecture develops the eigenfunction method for that setting.